

## ERBENT, Turkmenistan

WMO: 386560

Lat: 39.317N

Lon: 58.600E

Elev: 87

StdP: 100.28

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.6      | -7.7 | -18.6                         | 0.7 | -8.5 | -14.7 | 1.1 | -4.5 | 7.6                   | 5.7  | 6.3 | 6.5  | 1.6                   | 90   | 0.449 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 42.3            | 21.2 | 40.9 | 21.0 | 39.6 | 20.5 | 24.6                | 37.6 | 23.3 | 36.4 | 22.1 | 36.2 | 2.3                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.0                            | 14.9 | 34.3 | 18.5 | 13.5 | 32.7 | 16.7 | 12.0 | 31.0 | 75.1          | 37.8 | 69.7 | 36.1 | 65.0 | 36.0 | 34.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 6.7           | 5.9  | 5.0 | -14.3                      | 44.5 | 5.6                | 1.3 | -18.3                                              | 45.5 | -21.5      | 46.3 | -24.7      | 47.0 | -28.7      | 48.0 |    |    |
| (5)               |      |     | -14.9                      | 25.8 | 5.3                | 3.0 | -18.7                                              | 27.9 | -21.8      | 29.6 | -24.8      | 31.3 | -28.7      | 33.5 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.6   | 2.6   | 4.6   | 11.5  | 18.1  | 25.2  | 30.3  | 32.6  | 30.9  | 24.5  | 16.9  | 9.0   | 3.6  | (6)  |
| (7)  |                                                                                  | DBStd       | 11.56  | 5.11  | 5.65  | 4.96  | 5.05  | 3.66  | 3.09  | 2.74  | 3.06  | 3.78  | 4.73  | 5.16  | 4.53 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 725    | 233   | 162   | 41    | 4     | 0     | 0     | 0     | 0     | 0     | 6     | 76    | 204  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1983   | 488   | 386   | 219   | 64    | 2     | 0     | 0     | 0     | 3     | 82    | 282   | 457  | (9)  |
| (10) |                                                                                  | CDD10.0     | 3482   | 2     | 9     | 87    | 248   | 471   | 609   | 702   | 647   | 437   | 221   | 46    | 5    | (10) |
| (11) |                                                                                  | CDD18.3     | 1699   | 0     | 0     | 6     | 58    | 214   | 359   | 443   | 388   | 189   | 38    | 2     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 25052  | 0     | 1     | 92    | 644   | 2661  | 5538  | 7331  | 6022  | 2313  | 428   | 22    | 1    | (12) |
| (13) |                                                                                  | CDH26.7     | 15420  | 0     | 0     | 27    | 261   | 1379  | 3515  | 4992  | 3934  | 1166  | 141   | 5     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.0    | 1.9   | 2.2   | 2.4   | 2.4   | 2.2   | 2.2   | 2.0   | 1.8   | 1.9   | 1.8   | 1.7   | 1.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 106    | 13    | 10    | 18    | 21    | 13    | 2     | 2     | 1     | 2     | 6     | 10    | 13   | (15) |
| (16) |                                                                                  | PrecMax     | 202    | 36    | 39    | 73    | 74    | 60    | 9     | 19    | 5     | 15    | 27    | 30    | 47   | (16) |
| (17) |                                                                                  | PrecMin     | 61     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 33     | 9     | 9     | 16    | 19    | 15    | 2     | 4     | 1     | 3     | 6     | 8     | 11   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 18.0  | 21.3  | 30.2  | 36.0  | 39.2  | 42.7  | 44.7  | 43.7  | 39.6  | 33.7  | 26.4  | 19.3 | (19) |
| (20) |                                                                                  |             | MCWB   | 9.0   | 12.4  | 14.6  | 18.2  | 19.4  | 22.2  | 22.0  | 21.2  | 19.8  | 17.6  | 15.2  | 10.4 | (20) |
| (21) |                                                                                  | 2%          | DB     | 14.0  | 17.6  | 25.7  | 32.3  | 36.8  | 40.9  | 42.9  | 41.7  | 36.8  | 30.2  | 21.7  | 14.9 | (21) |
| (22) |                                                                                  |             | MCWB   | 7.7   | 9.8   | 13.5  | 16.7  | 19.3  | 21.1  | 21.2  | 20.9  | 18.9  | 16.0  | 12.4  | 8.7  | (22) |
| (23) |                                                                                  | 5%          | DB     | 11.2  | 14.8  | 22.5  | 29.7  | 35.0  | 39.5  | 41.4  | 40.2  | 34.8  | 27.9  | 19.0  | 12.0 | (23) |
| (24) |                                                                                  |             | MCWB   | 6.5   | 8.3   | 12.3  | 15.8  | 18.4  | 20.2  | 21.0  | 20.5  | 18.6  | 15.1  | 11.2  | 7.2  | (24) |
| (25) |                                                                                  | 10%         | DB     | 9.0   | 12.2  | 19.7  | 27.1  | 33.2  | 38.1  | 40.0  | 38.7  | 32.7  | 25.5  | 16.7  | 9.7  | (25) |
| (26) |                                                                                  |             | MCWB   | 5.3   | 7.2   | 11.2  | 14.9  | 17.9  | 19.7  | 21.0  | 20.0  | 17.5  | 13.9  | 10.2  | 5.8  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.4  | 12.7  | 15.5  | 19.3  | 22.6  | 25.4  | 26.6  | 25.8  | 24.2  | 19.4  | 15.7  | 11.4 | (27) |
| (28) |                                                                                  |             | MCD B  | 15.1  | 19.6  | 27.2  | 31.6  | 34.0  | 39.3  | 39.5  | 37.3  | 35.7  | 29.3  | 25.6  | 18.0 | (28) |
| (29) |                                                                                  | 2%          | WB     | 8.4   | 10.4  | 14.1  | 17.9  | 20.7  | 23.6  | 25.0  | 23.6  | 21.1  | 17.1  | 13.3  | 9.2  | (29) |
| (30) |                                                                                  |             | MCD B  | 13.1  | 16.3  | 23.8  | 29.7  | 33.2  | 36.9  | 38.3  | 36.2  | 32.8  | 27.9  | 19.9  | 14.3 | (30) |
| (31) |                                                                                  | 5%          | WB     | 6.8   | 8.8   | 13.0  | 16.6  | 19.6  | 22.4  | 23.7  | 22.1  | 19.5  | 15.8  | 11.9  | 7.5  | (31) |
| (32) |                                                                                  |             | MCD B  | 11.1  | 13.9  | 21.3  | 27.8  | 32.1  | 35.9  | 37.1  | 36.8  | 31.6  | 25.9  | 18.2  | 11.3 | (32) |
| (33) |                                                                                  | 10%         | WB     | 5.4   | 7.5   | 11.7  | 15.4  | 18.7  | 21.1  | 22.3  | 20.8  | 18.3  | 14.5  | 10.5  | 6.3  | (33) |
| (34) |                                                                                  |             | MCD B  | 8.8   | 12.0  | 18.8  | 25.4  | 31.0  | 35.0  | 36.6  | 37.1  | 31.0  | 24.1  | 16.2  | 9.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.7   | 9.6   | 11.1  | 11.4  | 12.5  | 12.8  | 12.6  | 13.5  | 13.5  | 12.9  | 10.4  | 8.3  | (35) |
| (36) |                                                                                  |             | MCD BR | 12.7  | 14.0  | 15.7  | 15.2  | 14.8  | 14.7  | 14.2  | 15.1  | 15.6  | 15.6  | 13.8  | 12.4 | (36) |
| (37) |                                                                                  |             | MCWBR  | 7.5   | 7.8   | 6.8   | 6.2   | 6.1   | 5.9   | 5.6   | 6.4   | 7.1   | 7.7   | 7.1   | 7.6  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 11.9  | 13.0  | 14.2  | 13.7  | 13.0  | 12.8  | 12.0  | 13.3  | 13.6  | 14.1  | 12.6  | 11.2 | (38) |
| (39) |                                                                                  |             | MCWBR  | 7.2   | 7.6   | 6.5   | 6.2   | 6.2   | 6.7   | 6.7   | 7.3   | 7.4   | 7.6   | 7.1   | 7.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.408  | 0.453 | 0.516 | 0.566 | 0.555 | 0.563 | 0.564 | 0.516 | 0.487 | 0.486 | 0.440 | 0.407 | (40) |      |
| (41) |                                                                                  | taud        | 1.998  | 1.881 | 1.775 | 1.719 | 1.777 | 1.761 | 1.770 | 1.854 | 1.891 | 1.871 | 1.958 | 1.997 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 710    | 732   | 731   | 727   | 748   | 740   | 735   | 759   | 749   | 689   | 665   | 672   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 127    | 164   | 202   | 228   | 219   | 223   | 219   | 196   | 178   | 163   | 132   | 119   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.12   | 3.00  | 4.28  | 5.62  | 6.86  | 7.89  | 7.64  | 7.05  | 5.76  | 3.98  | 2.62  | 1.89  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.20   | 0.32  | 0.26  | 0.45  | 0.32  | 0.22  | 0.24  | 0.25  | 0.17  | 0.25  | 0.19  | 0.25  | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) |  | N/A   | N/A     | N/A    | +0.64   | N/A   | N/A   | N/A         | N/A     | N/A     | +82     |

Nomenclature: See separate page